

MetaboCensoR Tutorial

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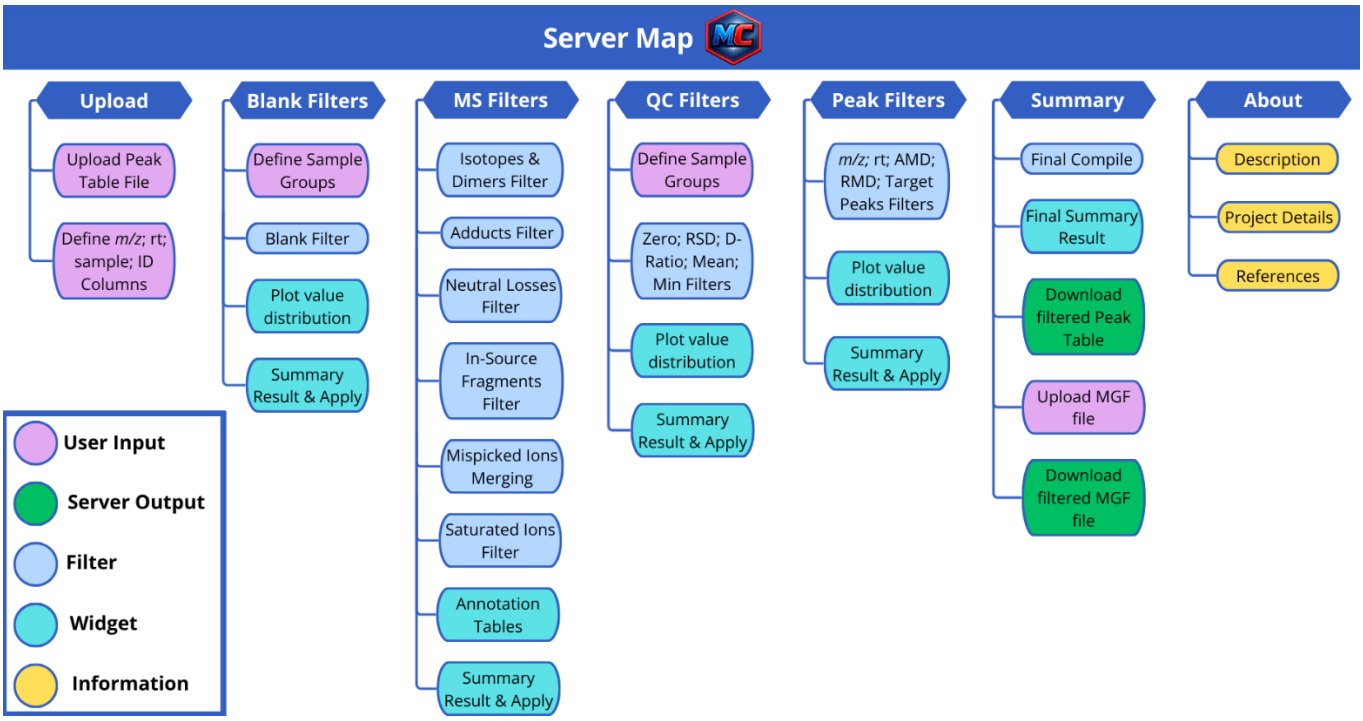
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General Information

This tutorial aims at guiding the user to process data LC-MS data through Shiny GUI interface of MetaboCensoR. MetaboCensoR is an input-versatile Shiny application and local R package for analyte-centric peak table filtering. The workflow integrates four complementary modules for blank filtering, redundant ion-species filtering, quality-control filtering, and peak-based filtering. MetaboCensoR also provides interactive threshold optimization, exportable annotation tables, and synchronized filtering of associated .mgf files.

An overview of MetaboCensoR is presented below. The workflow is organized into six functional tabs that collectively support peak table curation, parameter optimization, and export of filtered results. The output of the app is a filtered peak table and, optionally, an MGF spectra file. Only the peak table upload is required for full workflow execution and final output generation.

License: GPL-3.0
Contact: plyushchenko.ivan@gmail.com
Project Repository: github.com/plyush1993/MetaboCensoR
Citation: Plyushchenko, I. V., & Luzzatto-Knaan, T. (2026). bioRxiv, 2026-07.



Running & Installation

Shiny Web Deployment is available by the link:

plyush1993.shinyapps.io/metabocensor/

Local installation of R package:

```
if (!require("BiocManager", quietly = TRUE)) {
    install.packages("BiocManager")
}
if (!require("remotes", quietly = TRUE)) {
    install.packages("remotes")
}
remotes::install_github("plyush1993/MetaboCensoR", INSTALL_opts = "--no-multiarch")
```

or

```
if (!requireNamespace("pak", quietly = TRUE)) install.packages("pak")
pak::pak("plyush1993/MetaboCensoR")
```

Run Locally:

```
MetaboCensoR::run_metabocensor()
```

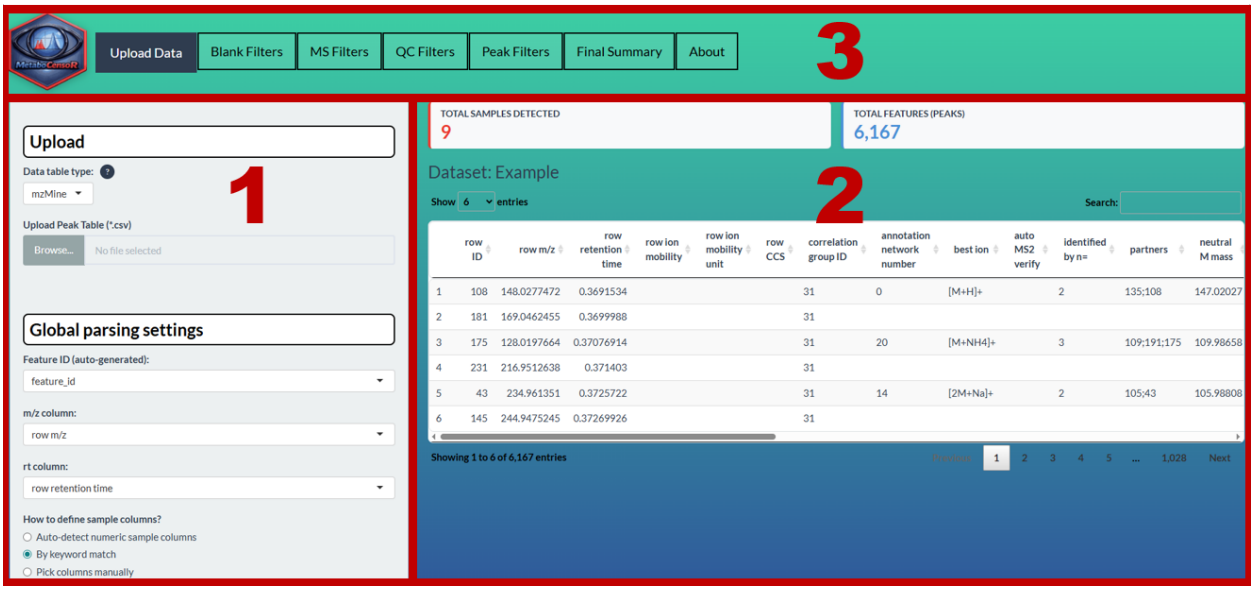
Input Requirements

The application accepts peak tables in ***.csv*** format and MS1/MS2 mass spectral data in ***.mgf*** format, both of which can be directly exported from dedicated metabolomics processing software (xcms, MZmine, MS-DIAL). Make sure that your input peak table is aligned with data formats. Examples of acceptable formats are available here: Input_Examples.

Important note

The Shiny server has limited memory capacity (**1 GB**). Computationally intensive steps (like MS filters) or large MGF files may cause the app to stack or disconnect. For large datasets or memory-intensive analyses, we recommend activating the "**Limit table previews**" option and running *MetaboCensoR* locally

Overview of Interface



GUI is organized into 3 main panels:

1. The Sidebar Panel (Control Panel)

- This is the primary area for user interaction. It houses the input controls where users upload data, configure global settings, define parameters, and execute filtering steps.

2. The Main Panel (Display Area)

- This is where the application renders the output and displays interactive data tables, statistical plots, and provides summary metrics.

3**. The Navigation Bar (Tab Menu)**

- This allows users to navigate linearly through the sequential steps of the data processing pipeline, from data upload to final summary and export.

Contextual assistance is embedded throughout the interface. Users can access detailed parameter descriptions and recommendations by interacting with the tooltip icons.

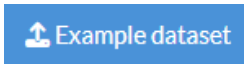
On the following pages, yellow numbered circles indicate the recommended operational order within each interface tab in *Control Panel*.

Red numbered squares describe widgets in *Display Area*.

Implementation Details

All filter groups are applied sequentially in the order shown in the tab panel: Blank Filters → MS Filters → QC Filters → Peak Filters. Changes made on later tabs do not affect the results of previous tabs. To change the order of applied filters (e.g., 'Peak Filters' before 'Blank Filters'), run the application sequentially several times, using the output peak table from one run as the input for the next.

Quick Start

Press button 

in “Upload data” tab to upload attached example dataset and get access to the MetaboCensoR functionality. If you use this example, run the Blank filter first to save memory for further steps. You can keep all parameters by default.

Briefly, it is the LC-MS plant profiling dataset, which is described in *Houriet et al*. Briefly, methanol extracts from the plant ashwagandha [*Withania somnifera (L.) Dunal*] together with blanks were analyzed on RP column in DDA positive mode on Thermo Q-Exactive Plus Orbitrap. Raw mzML files were then processed in mzMine in default pre-settings for UPLC-DDA to obtain resulted peak table.

Tab “Upload Data”: Control Panel



1. Pick the right table type and upload peak table

MetaboCensoR supports mzMine, xcms, MS-DIAL, and Generic input tables. If the type is wrong, the app throws a parsing error.

2. Specify peak characteristic columns

Use the selectors to check/select 3 columns:

- Feature ID
- m/z
- rt

We recommend to start with default column names.

3. Specify sample intensity columns

Use auto-detect, keyword matching, or manual column selection.

4. Limit table previews

For large datasets we recommend activating the “**Limit table previews**” option

5. Clear dataset

Reset all calculations.

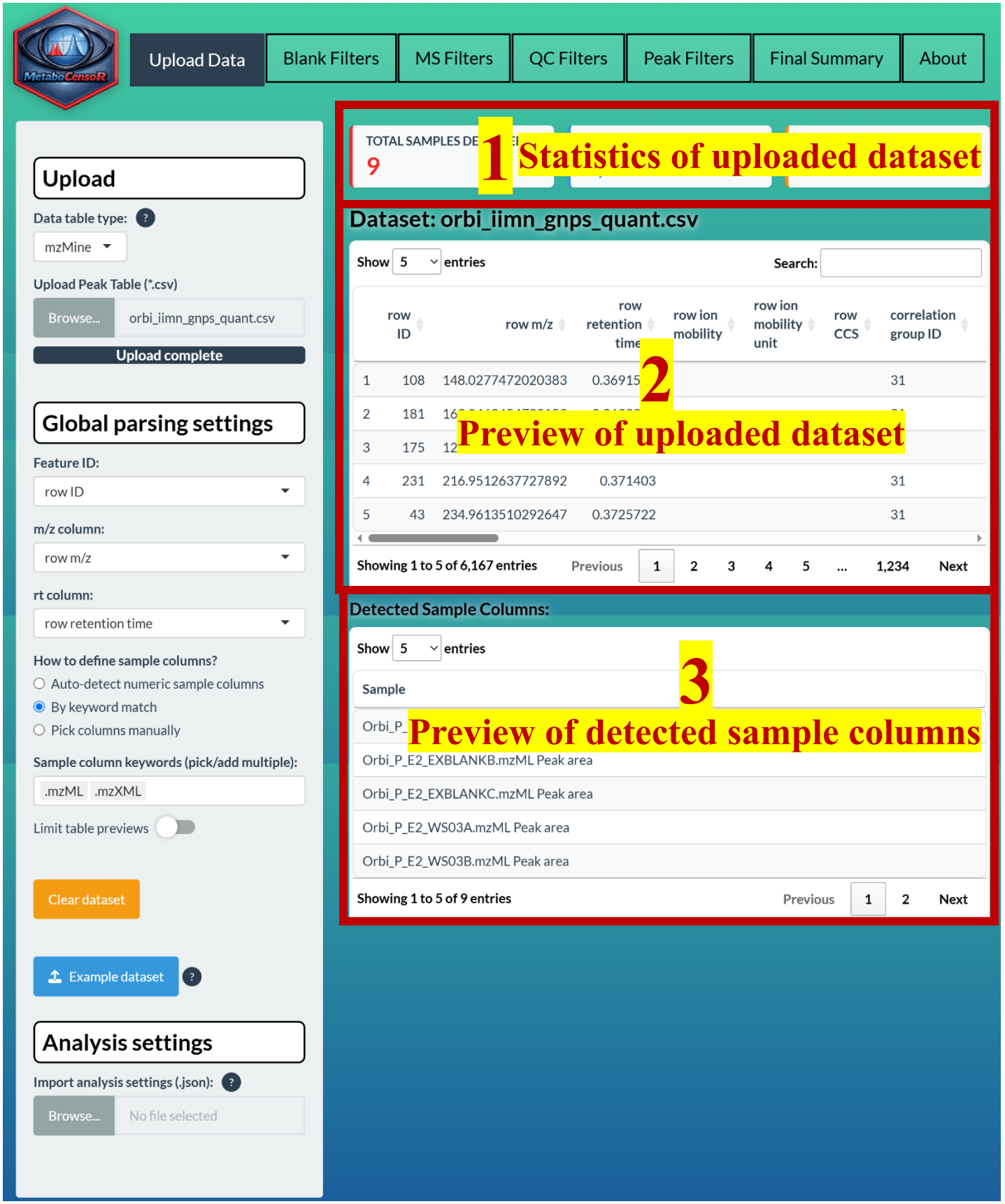
6. Example dataset shortcut

The Example dataset button loads the built-in example.

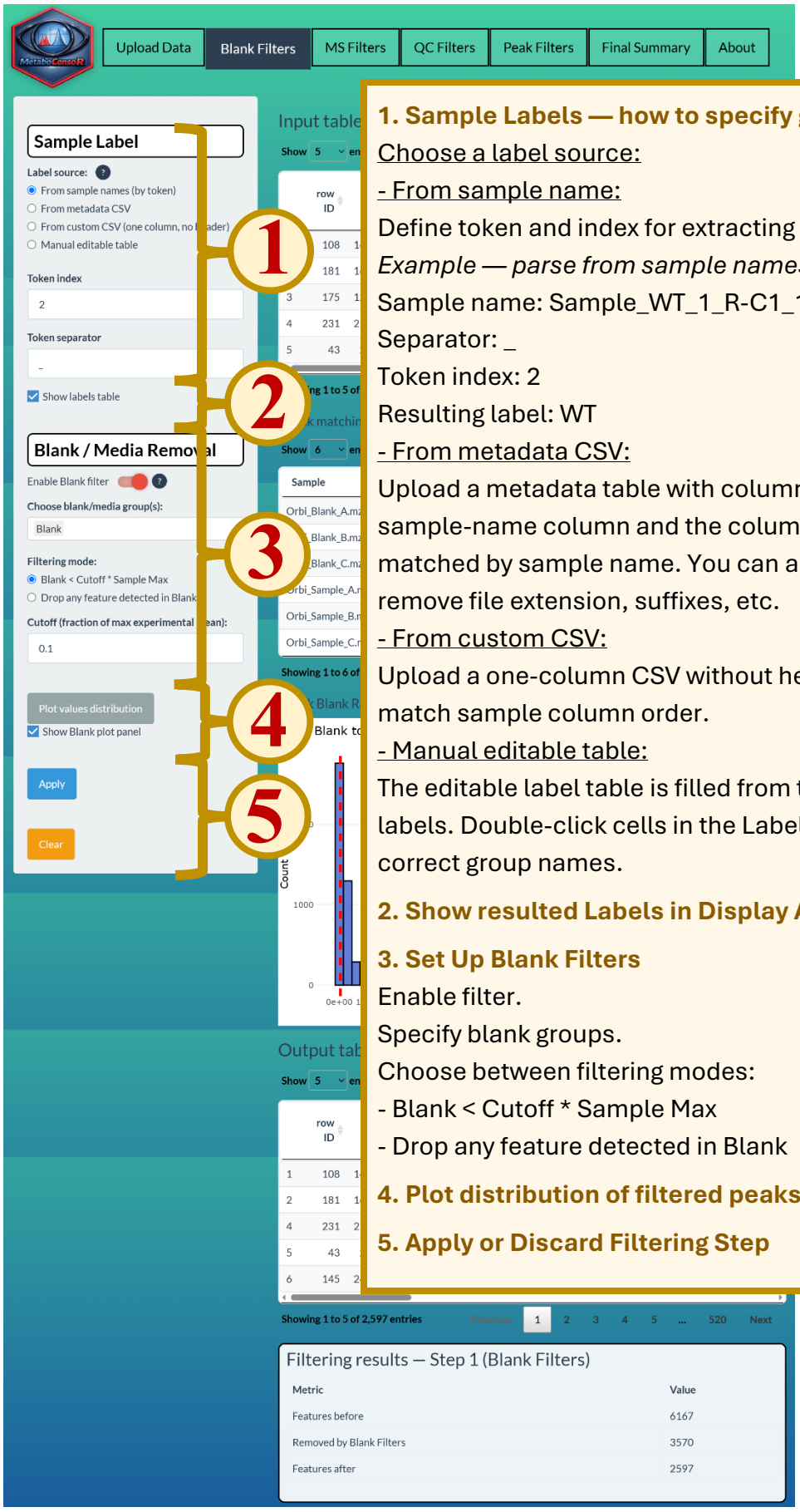
7. Upload analysis settings

Upload JSON to restore settings.

Tab “Upload Data”: Display Area



Tab “Blank Filters": Control Panel



1. Sample Labels — how to specify groups

Choose a label source:

- From sample name:

Define token and index for extracting labels from sample names.

Example — parse from sample names

Sample name: Sample_WT_1_R-C1_1_5037.mzML Peak area

Separator: _

Token index: 2

Resulting label: WT

- From metadata CSV:

Upload a metadata table with column names, choose the sample-name column and the column to use as Label. Rows are matched by sample name. You can also clean sample name to remove file extension, suffixes, etc.

- From custom CSV:

Upload a one-column CSV without header. Label order must match sample column order.

- Manual editable table:

The editable label table is filled from the current token-based labels. Double-click cells in the Label column to manually correct group names.

2. Show resulted Labels in Display Area

3. Set Up Blank Filters

Enable filter.

Specify blank groups.

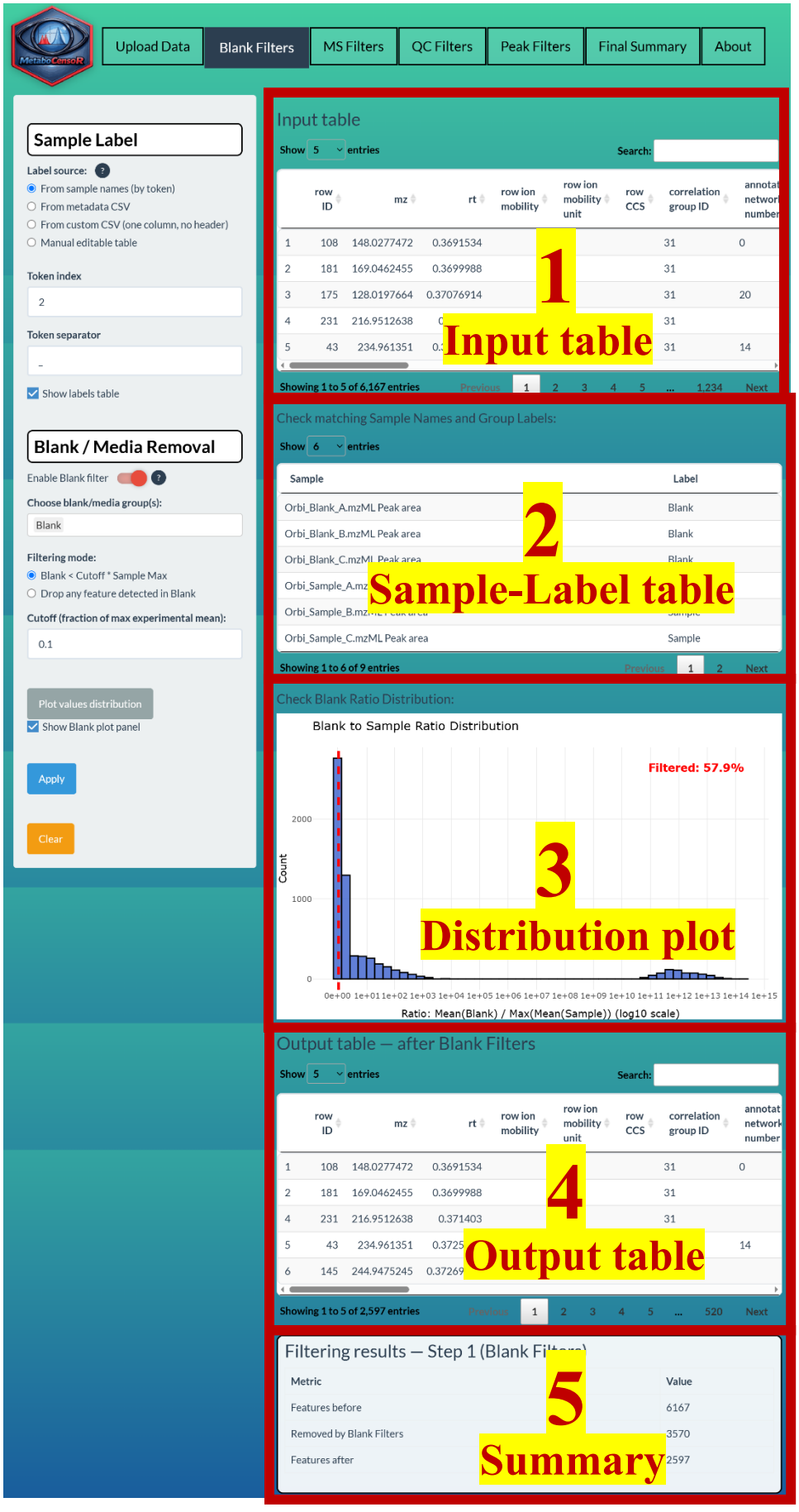
Choose between filtering modes:

- Blank < Cutoff * Sample Max
- Drop any feature detected in Blank

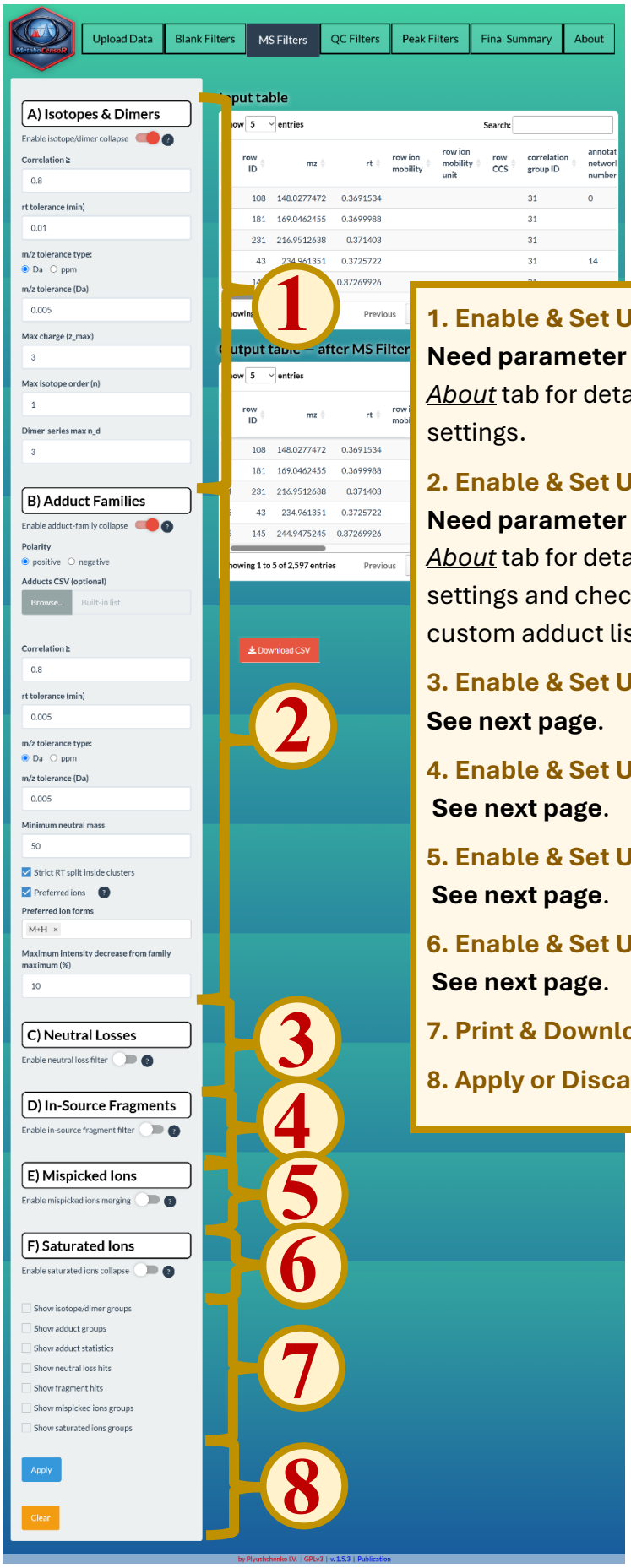
4. Plot distribution of filtered peaks

5. Apply or Discard Filtering Step

Tab “Blank Filters": Display Area



Tab “MS Filters”: Control Panel



1. Enable & Set Up Isotopes Filtering

Need parameter help? Click the tooltips or read the *About* tab for details. **Pro-Tip:** Start with the default settings.

2. Enable & Set Up Adducts Filtering

Need parameter help? Click the tooltips or read the *About* tab for details. **Pro-Tip:** Start with the default settings and check your polarity. You can upload custom adduct list in csv.

3. Enable & Set Up Neutral Losses Filtering

See next page.

4. Enable & Set Up In-Source Fragments Filtering

See next page.

5. Enable & Set Up Mispicked Ions Filtering

See next page.

6. Enable & Set Up Saturated Ions Filtering

See next page.

7. Print & Download annotation tables

8. Apply or Discard Filtering Step



3. Enable & Set Up Neutral Losses Filtering

Need parameter help? Click the tooltips or read the *About* tab for details. **Pro-Tip:** Use if only MS1 is available.

4. Enable & Set Up In-Source Fragments Filtering

Need parameter help? Click the tooltips or read the *About* tab for details. **Pro-Tip:** Use with "**Require MS2 fragment match**" option if MS2 data as is available. For this upload MGF file corresponding to the peak table.

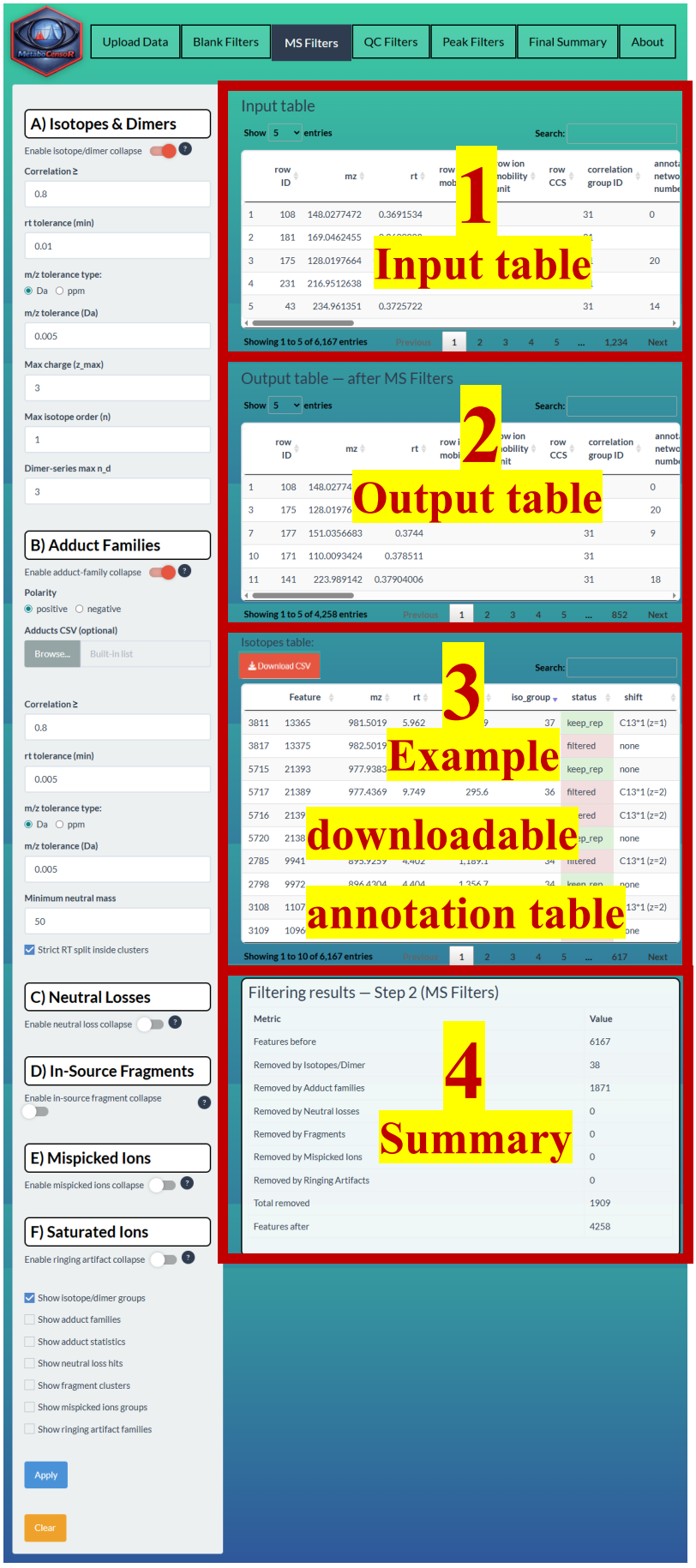
5. Enable & Set Up Mispicked Ions Filtering

Need parameter help? Click the tooltips or read the *About* tab for details.

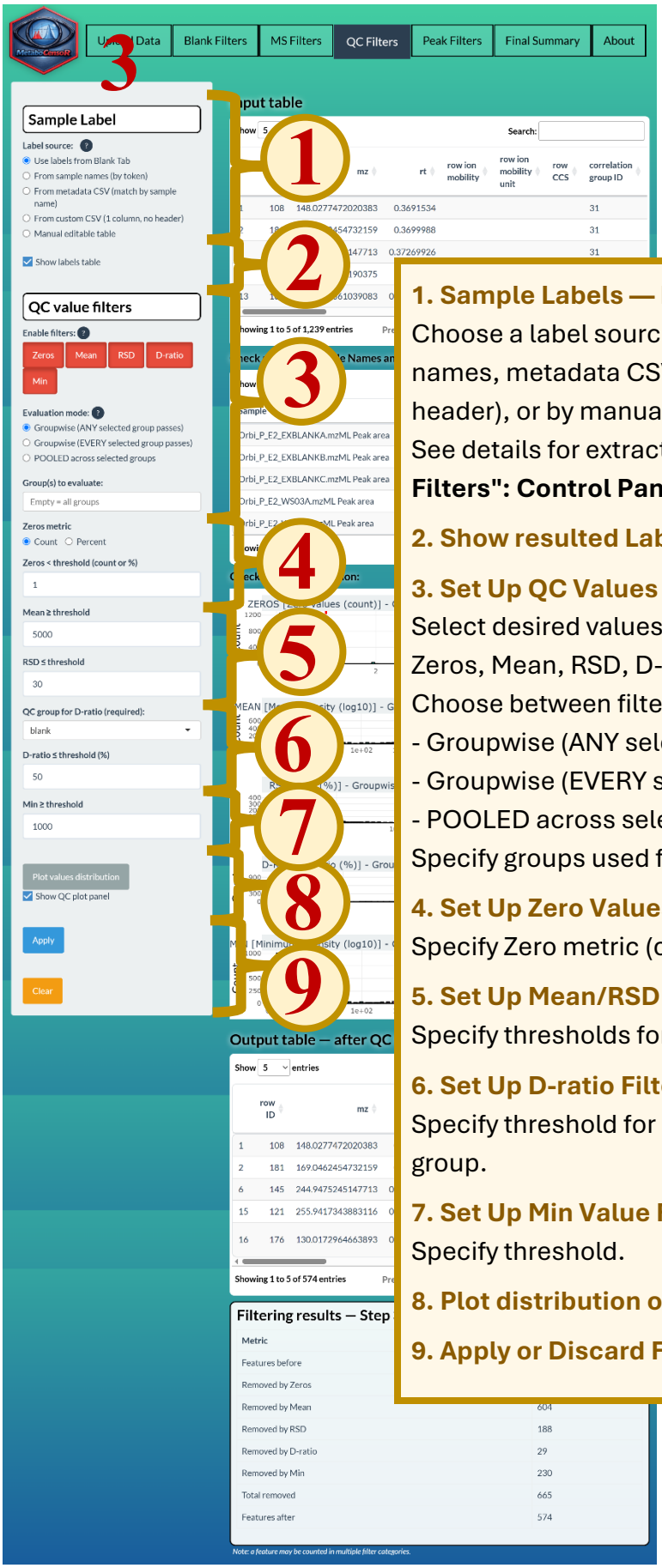
6. Enable & Set Up Saturated Ions Filtering

Need parameter help? Click the tooltips or read the *About* tab for details.

Tab “MS Filters”: Display Area



Tab “QC Filters”: Control Panel



1. Sample Labels — how to specify groups

Choose a label source: from *Blank Filters* tab, sample names, metadata CSV, custom CSV (one column, no header), or by manual editing.

See details for extracting Labels on the Page **Tab “Blank Filters": Control Panel**.

2. Show resulted Labels in Display Area

3. Set Up QC Values Filters

Select desired values for filtering:

Zeros, Mean, RSD, D-ratio or Min.

Choose between filtering modes:

- Groupwise (ANY selected group passes)
- Groupwise (EVERY selected group passes)
- POOLED across selected groups

Specify groups used for evaluation.

4. Set Up Zero Values Filters

Specify Zero metric (counts or %) and threshold.

5. Set Up Mean/RSD Values Filters

Specify thresholds for corresponding value.

6. Set Up D-ratio Filter

Specify threshold for corresponding value and select QC group.

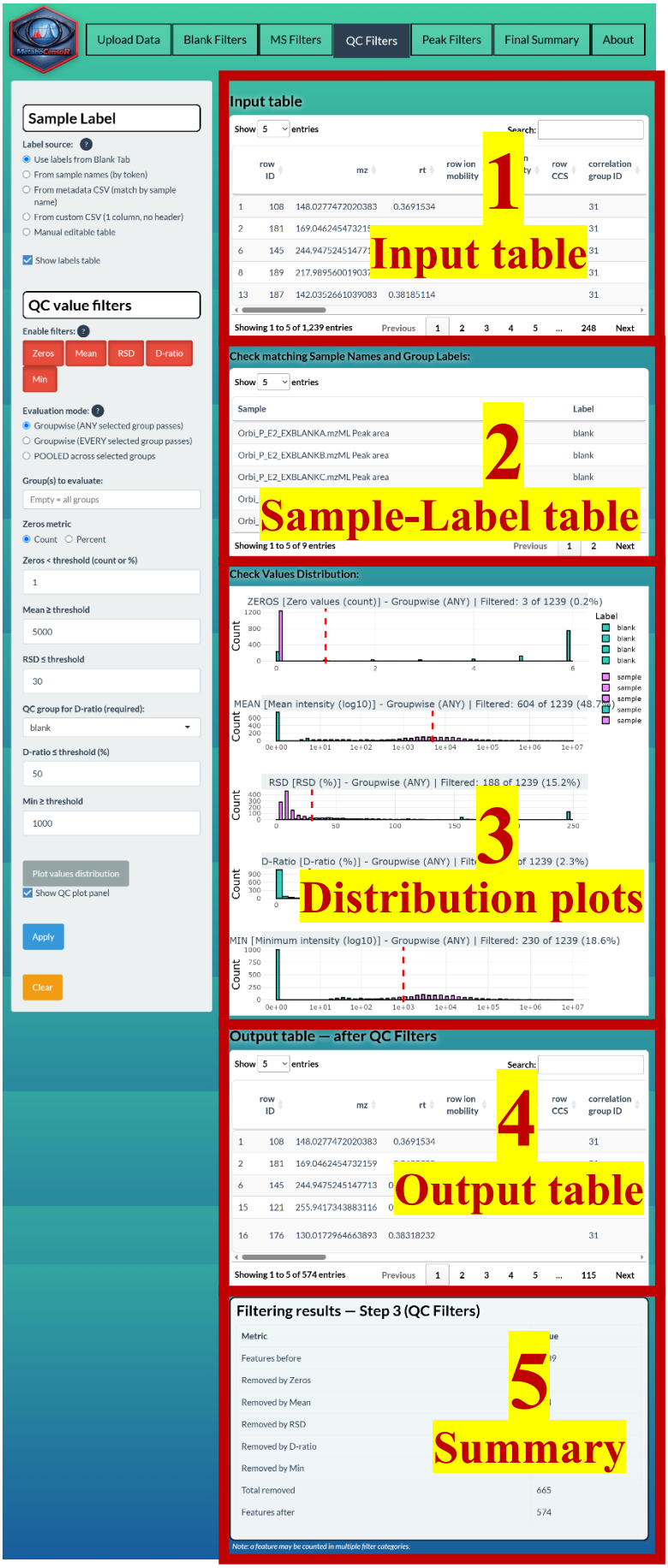
7. Set Up Min Value Filter

Specify threshold.

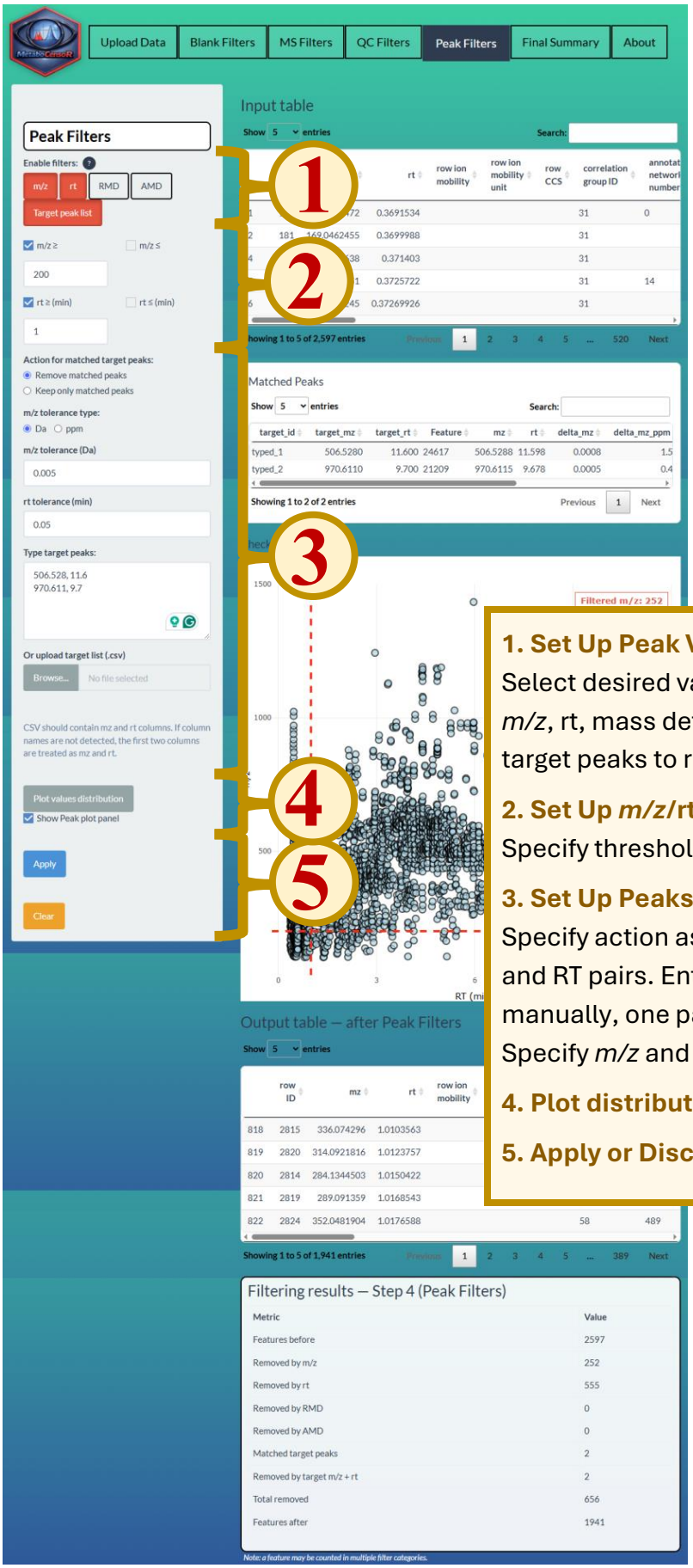
8. Plot distribution of selected values

9. Apply or Discard Filtering Step

Tab “QC Filters”: Display Area



Tab “Peak Filters”: Control Panel



1. Set Up Peak Values Filters

Select desired values for filtering:
m/z, rt, mass defect (RMD, or AMD), or a list of target peaks to remove.

2. Set Up *m/z*/rt/AMD/RMD Values Filters

Specify threshold ranges for corresponding value.

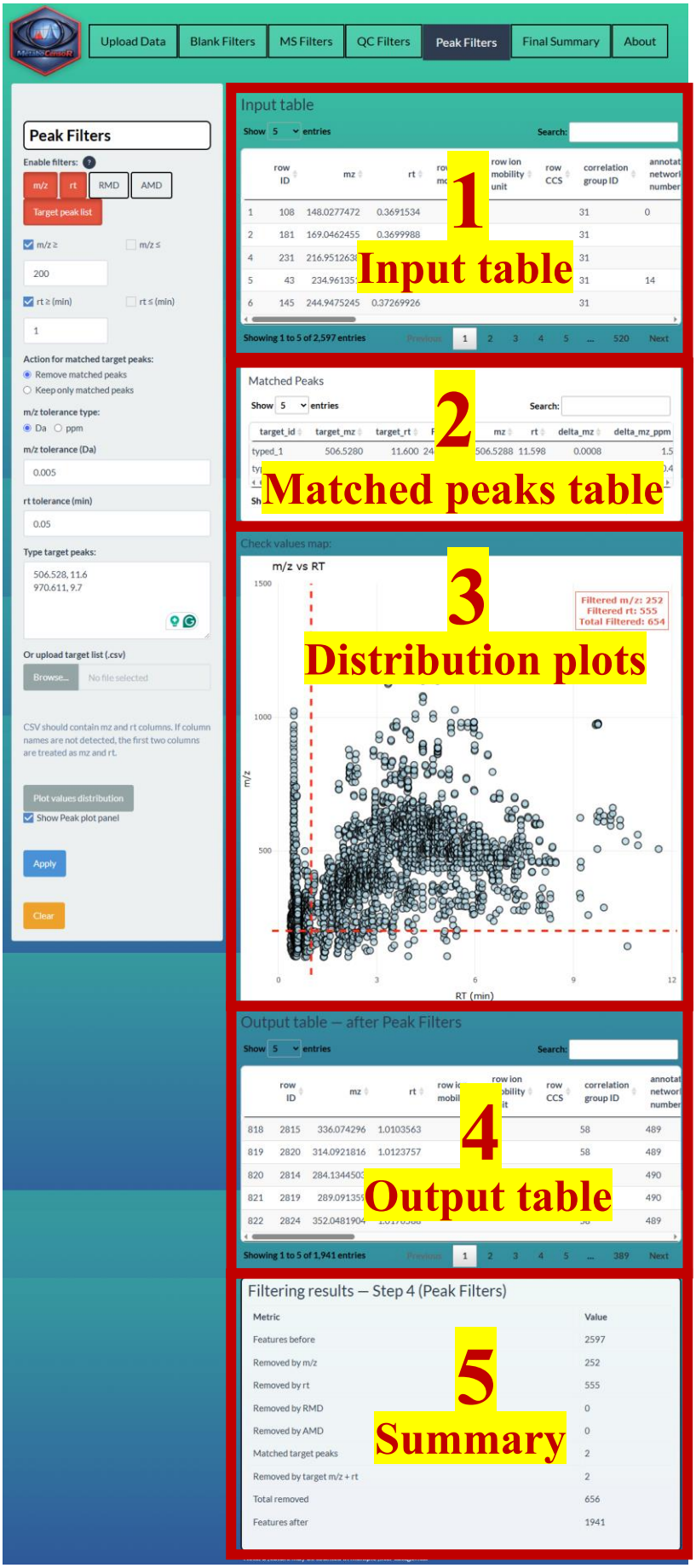
3. Set Up Peaks to Remove/Keep

Specify action as keep or remove; provide target *m/z* and RT pairs. Enter comma-separated values manually, one pair per line, or upload a CSV file. Specify *m/z* and RT tolerances for matching.

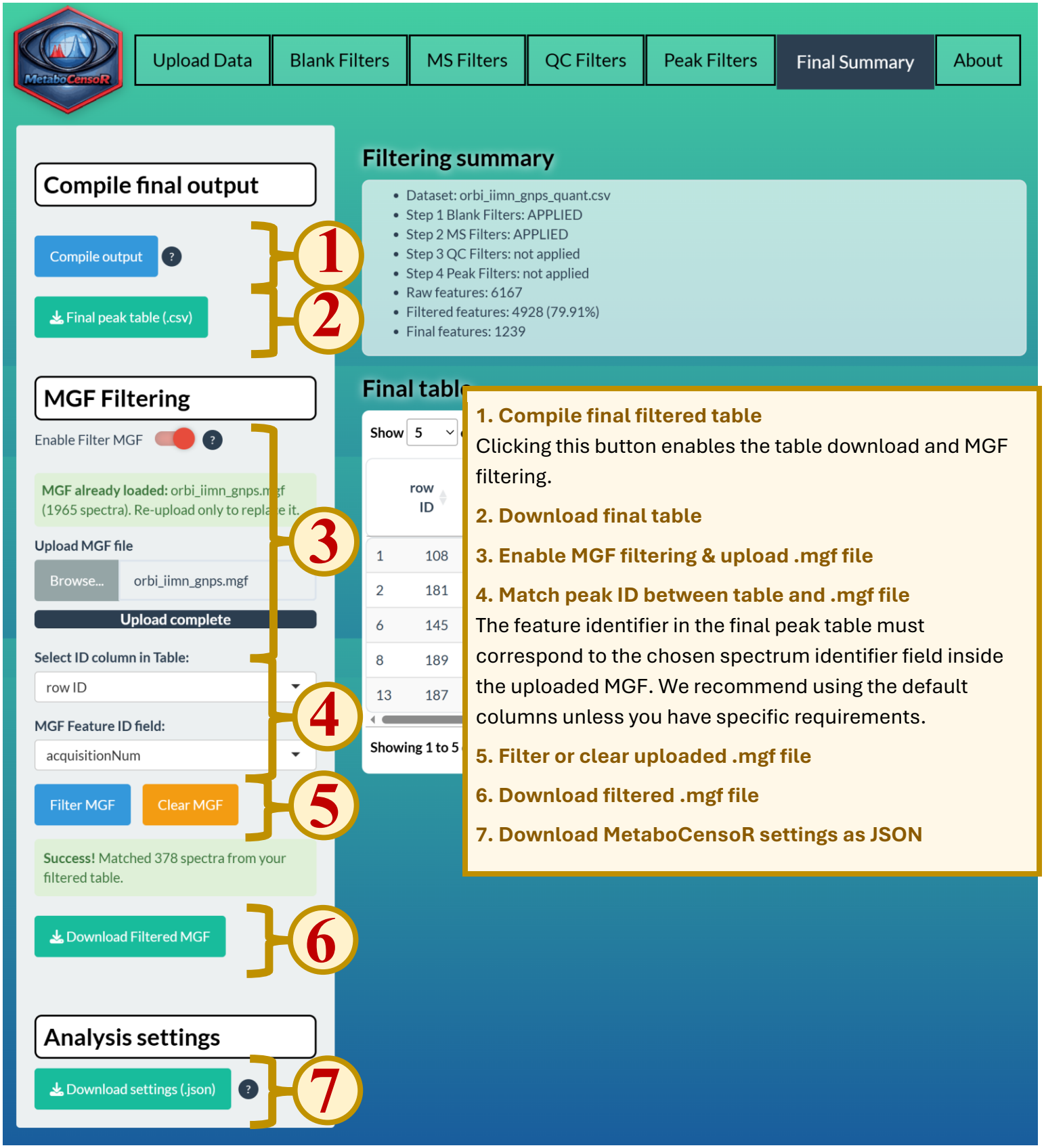
4. Plot distribution of selected values

5. Apply or Discard Filtering Step

Tab “Peak Filters”: Display Area



Tab “Final Summary”: Control Panel



1. Compile final filtered table
Clicking this button enables the table download and MGF filtering.

2. Download final table

3. Enable MGF filtering & upload .mgf file

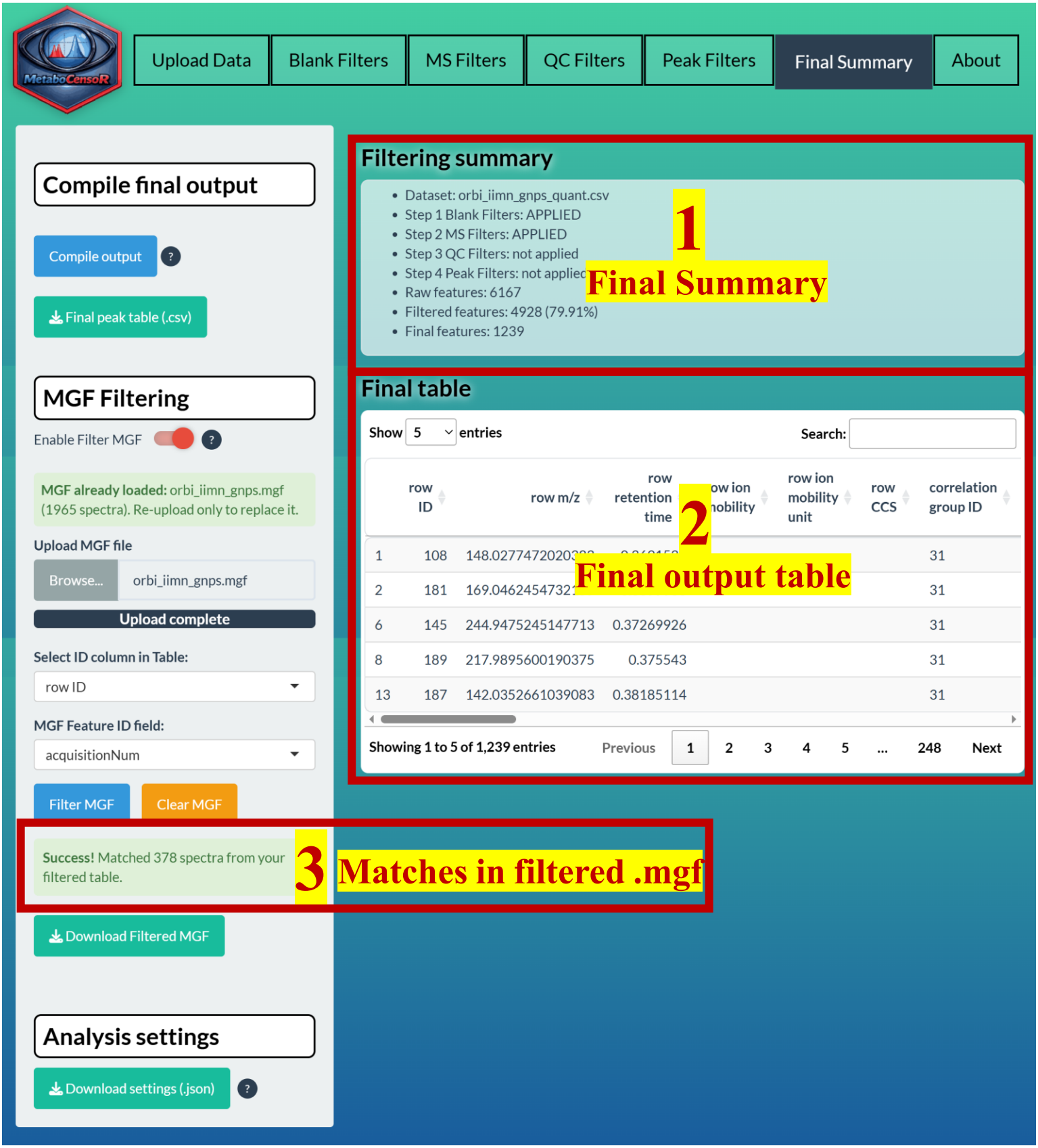
4. Match peak ID between table and .mgf file
The feature identifier in the final peak table must correspond to the chosen spectrum identifier field inside the uploaded MGF. We recommend using the default columns unless you have specific requirements.

5. Filter or clear uploaded .mgf file

6. Download filtered .mgf file

7. Download MetaboCensoR settings as JSON

Tab “Final Summary”: Display Area



Tab “About” Overview

